



Bay Area Gentrification Risk Prediction by Tract

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ABSTRACT

Gentrification is a process where **wealthier individuals and real estate developers move into historically disinvested, lower-income neighborhoods**. This drives up property values, rents, and taxes, and can drive out original residents who can no longer afford the rising costs.

Our goal was to build a model that not only flags high-risk neighborhoods but also explains why each tract is at risk. **For every census tract, the model produces both a relative risk score and a breakdown of the specific features driving that score.** This gives policymakers a tract-specific picture of the pressures at play, enabling interventions that are tailored to the needs of each individual community.

DATA SOURCES

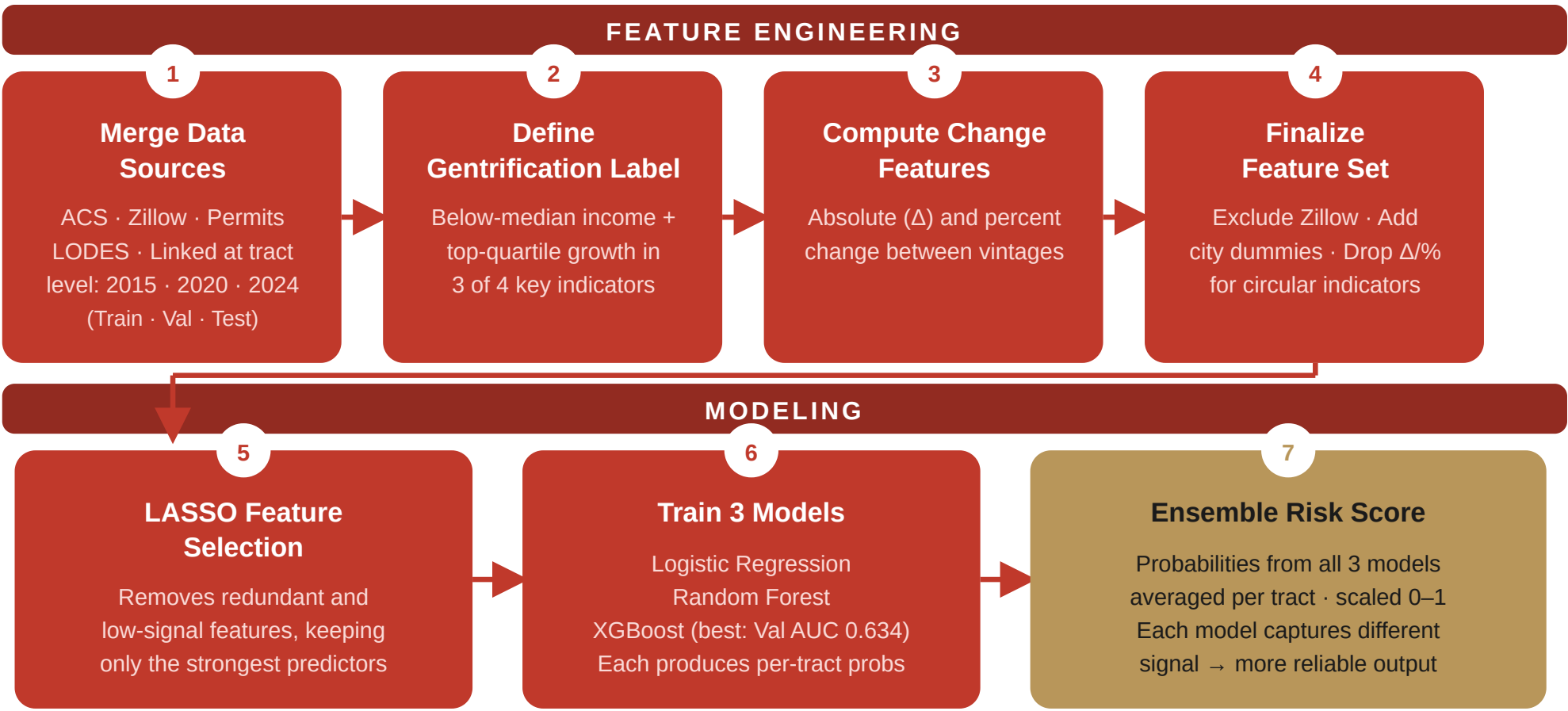
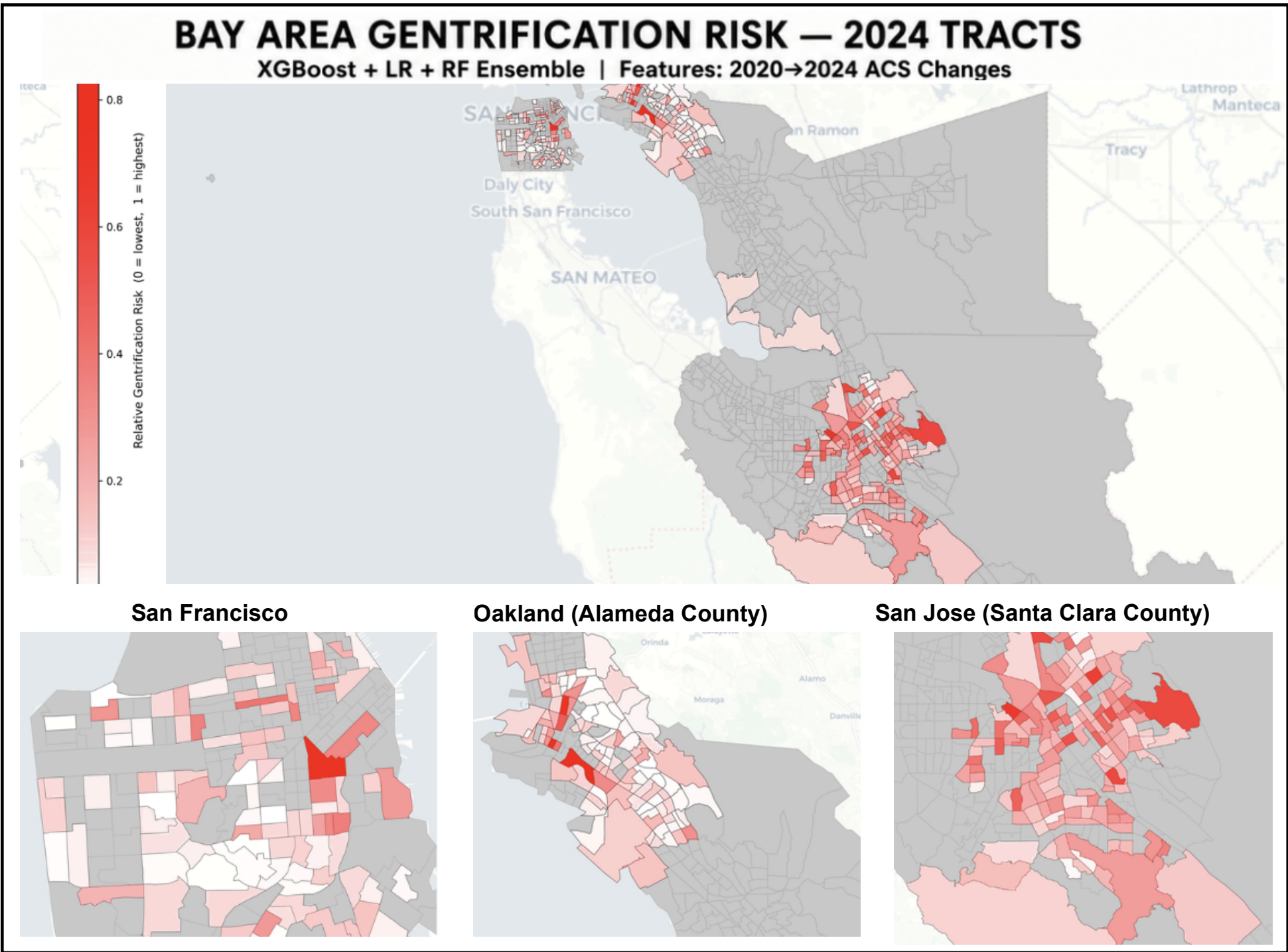
- For this project, we merged data at the tract level to create a diverse feature set, including:
- **Census data from the American Community Survey (ACS)** of 5-year rolling averages on income, rent, home value, college attainment, population statistics, poverty rates, and vacancy rates, among others.
 - **Longitudinal Employer-Household Dynamics (LODES)** data on job types per household.
 - **Building permit records** for counts of residential and commercial construction activity.
 - **The Zillow Home Value Index** data provides period-level housing appreciation figures.

GENTRIFICATION DEFINITION

- Gentrification is identified in two steps:**
1. The neighborhood must be **relatively low-income** to begin with (below the city median)
 2. It must demonstrate rapid change, **landing in the top quartile of all tracts for growth in at least 3 of 4 areas:**
 - a. Household Income
 - b. Home Values
 - c. Rent
 - d. College Education Percentage

RESEARCH QUESTION

Can neighborhood-level data from 2011-2024 (including permit activity, business density, and demographic composition) predict which Bay Area census tracts are at the highest risk of gentrification in the near future? Which features lead to a specific tract being labeled as high-risk?



BEST PREDICTORS

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XGBoost's top SHAP predictors on 2024 data:

- Change in the African American population
- Overall population size
- Share of jobs in the food industry
- Share of houses built before 1940
- Share of population ages 25-34

MODEL PERFORMANCE

Model	Val AUC	CV AUC	CV Std
LASSO Log. Regression	0.587	0.693	±0.176
Random Forest	0.597	0.582	±0.261
★ XGBoost (selected)	0.634	0.793	±0.159
3-Model Ensemble	0.604	—	—

LIMITATIONS

- **Small Training Set:** With so few true positives, we used SMOTE to upweight these during training, which leads to higher variance but increases predictive accuracy.
- **Market Regime Shift:** By training on mostly pre-COVID data and testing on post-COVID transitions, we are looking at data with slightly different trends, affecting predictions.
- **ACS Rolling Average Overlap:** The validation and testing data are 5-year averages, but both include 2020 due to the delay in posting new information. No true data leakage exists due to averaging with 4 separate years of data.

POLICY IMPLICATIONS

- **Provides an early-warning system** that helps cities proactively target affordable housing preservation, tenant protections, community land trusts, and anti-displacement resources toward neighborhoods exhibiting early signs of gentrification.
- The model's impact depends on who uses it; policymakers may use risk scores to **mitigate displacement**, while investors/developers could use the same information to **guide speculation**, potentially creating a feedback loop that reinforces the gentrification the model predicts.